Artificial Intelligence

159.302

Teaching Team

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Computer Science

School of Mathematical and Computational Sciences



Topics for Discussion

- Pre-requisites
- Course Overview
- Learning Outcomes
- Texts and Course Material
- Assessment
- Course Schedule
- AI Demonstrations

*

Pre-requisites

- To take this course you must have passed 159.201 or 159.234. Knowledge of C/C++ is required to complete the assignments.

- Students without any background in C++ data structures and basic object-oriented programming are very likely to fail in all of the assignments.

My advice is not to take this course without satisfying the pre-requisites for this course.

Teaching Approach

- **Discussion of the Theoretical Framework**
- **Step-by-Step Algorithm Details**

Course Overview

- **Application to real-world problems**
- **Simulations** *(A Graphics Engine will be provided to make learning AI more fun)*

Learning Outcomes

On successful completion of the course, the students should be able to:

1. Describe key Artificial Intelligence techniques and problem-solving methods.
2. Design and implement computer programs to store and manipulate knowledge for Artificial Intelligence applications..
3. Describe and implement Artificial Intelligence techniques for a range of target problems.
4. Apply logic systems for inference and control problems.
5. Identify the advantages and disadvantages of applying various AI techniques for solving real-world problems.

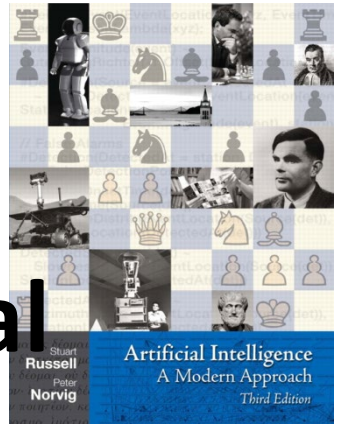
Texts and Course Material

Lecture slides, assignment instructions, codes, tutorial questions

- Stream web site

Recommended Reading

- Russell S. and Norvig P., Artificial Intelligence A Modern Approach, 3rd Ed, Prentice Hall 2009
  ISBN-13: 9780136042594



Other References

- MIT OpenCourseWare
- Neural Network and Fuzzy Logic Applications in C/C++ by Stephen T. Welstead
- Artificial Intelligence: Structures and Strategies for Complex Problem Solving by George F Luger

- **2 assignments: 40%**
- **Final Exam (3 hours): 60%** Bring your calculator

Assessment

To pass, students have to obtain a cumulative assessment score greater than or equal to **50%**.

Deadline: Deadlines for assignments will be given when assignments are distributed. You will be given 4 weeks to complete each assignment

Penalty: Late submissions (up to 1 week) will be penalised by 10% per day.

Assessment Description	Learning Outcomes Assessed					Contribution to Course Marks
	1	2	3	4	5	
Assignments		☐	☐	☐	☐	40.00%
Final Examination	☐	☐	☐	☐	☐	60.00%

Assessment

Assessment	Due Date / Deadline	Late Penalty	Course completion requirement	percentage
Assignment 1	30/Aug	-10% per day	n/a	20
Assignment 2	27/Sept	-10% per day	n/a	20
Final Exam	(exam timetable)		compulsory	60

Assessment

- Program solutions that do not compile or do not run using the required gcc compiler (see Stream website for directions for downloading and installing the required version) will get 0 mark.
- Late assignments will be penalised (-10% per day)
- Assignments may be completed in groups

  all members of the group should be named in the source file of each assignment, including the contribution of each member.

  All submitted assignments will have to be accompanied by some documentation of experimentations as well. The marks will be based on the results of those tests.

  Only one submission is required per group.

- There can be at most 3 members in a group.

Assessment

- Each group member will receive the same grade.
- Students in a team have the authority (in consultation with the lecturer) to "expel" any member who does not meet obligations .
- The collaboration is limited only to members within each group.
- It is a student responsibility to check their assignment marks and notify in writing any errors they might find no later than 10 days after the day the marks were made available.

Good News! (or bad news??)

- **Article: Is Google Cornering the Market on Deep Learning?**

A cutting-edge corner of science is being wooed by Silicon Valley, to the dismay of some academics. (January 29, 2014)

- How much are a dozen deep-learning researchers worth? Apparently, more than $400 million.
- This week, Google reportedly paid that much to acquire DeepMind Technologies, a startup based in London that had one of the biggest concentrations of researchers anywhere working on deep learning, a relatively new field of artificial intelligence research that aims to achieve tasks like recognizing faces in video or words in human speech see "Deep Learning").
- The acquisition, aimed at adding skilled experts rather than specific products, marks an acceleration in efforts by Google, Facebook, and other Internet firms to monopolize the biggest brains in artificial intelligence research.
- http://www.technologyreview.com/news/524026/is-google-cornering-the-market-on-deep-learning/

Week #	Topic
Week 1	Course Overview Intelligent Agents, Demonstrations, Introduction to Search: Background and Motivation, Problem Solving Paradigm, Classes of Search, Cost and Performance, Any-Path Search Algorithms
Week 2	Any Path Search Tutorial#1 Any Path Search Application: Robot Navigation - Tutorial#2 Optimal Search Algorithms
Week 3	Optimal Search – Part 1 - Tutorial#3 Optimality and Worst Case Complexity, Optimal Search – Part 2 - Tutorial#4 **Assignment #1** Instructions, Start-up codes
Week 4	Fuzzy Logic: Fuzzification, Inference Systems, Defuzzification, Fuzzy Logic Operators, Fuzzy Logic Engine Case Studies: Inverted Pendulum and Robot Navigation Fuzzy Logic - Tutorial#5 **Assignment #2** Instructions, Start-up codes
Week 5	Machine Learning: Perceptrons, Multi-layer Feed-Forward Networks Case Study: XOR Problem Backpropagation Learning - Tutorial#6

Week #	Topic
Week 6	Machine Learning: Backpropagation, Derivation of Formulas Case Study: Letter Recognition Multi-output network - Tutorial#7
	MID SEMESTER BREAK
Week 7	Constraint Satisfaction Problems (CSP): Motivation, Pure Backtracking (BT), Backtracking with Forward-Checking (BT-FC) CSP – Part 1 - Tutorial#8
Week 8	CSP: Dynamic Variable Ordering, Incremental Repair, GSAT, WalkSat CSP – Part 2 -Tutorial#9

Week #	Topic
Week 9	Games: Motivation, Board Games and Search, Min-Max, Alpha-Beta Pruning. Case Study: IBM Deep Blue, Games - Tutorial#10
Week 10	Logic: Motivation, Syntax, Semantics, Proof System, Sentences, Propositional Logic, Natural Deduction
Week 11	Logic Proof System: Propositional Resolution Logic – Tutorial#11
Week 12	Natural Language Processing: Technologies, Context-Free Grammar, Finite State Machine Parser Tips for the Finals
	STUDY BREAK
	EXAMS

How to approach your study?

Each week covers a specific set of concepts, terminologies and algorithms. You benefit if you do:

- Watch the video lectures.
- Read the relevant lectures and reference materials that refer to the topic presented on that week.
- Solve the tutorial questions provided for the topic. Some algorithms may require some mathematical background; therefore, you should solve the problem sets that accompany the topics, to check and see if you misunderstood something, or have questions.

How to approach your study?

- Complete the assignments on time. The lecture slides and start-up codes contain important clues that help with the completion of the assignments. Note that all assignment start-up codes were developed to run on a Windows 10 environment (as is the case with the machines at Massey's computer laboratories). Therefore, you should set-up a machine running the Windows operating system, if you want to write programs at home, for your assignments.
- It is very important to code, compile, debug and run your own programs.

Artificial Intelligence

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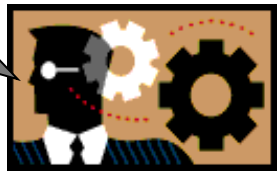
Sample Applications

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Motion Planning

Fast reactionary algorithm

How can we construct the intelligence for self-driving robots?

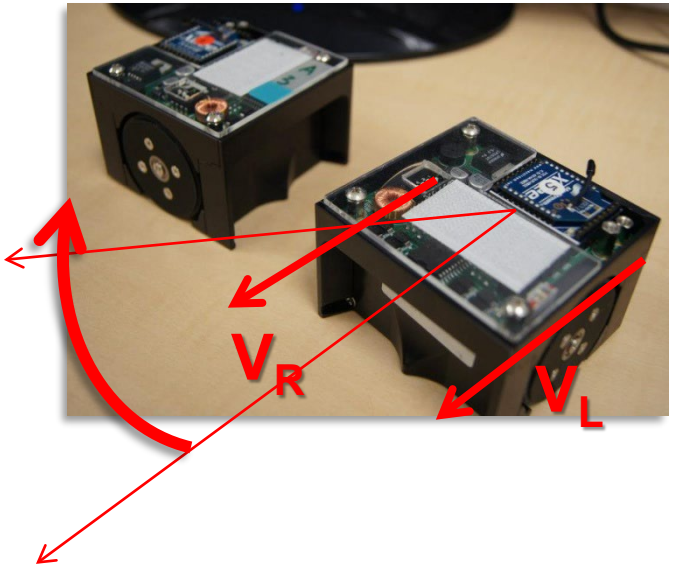


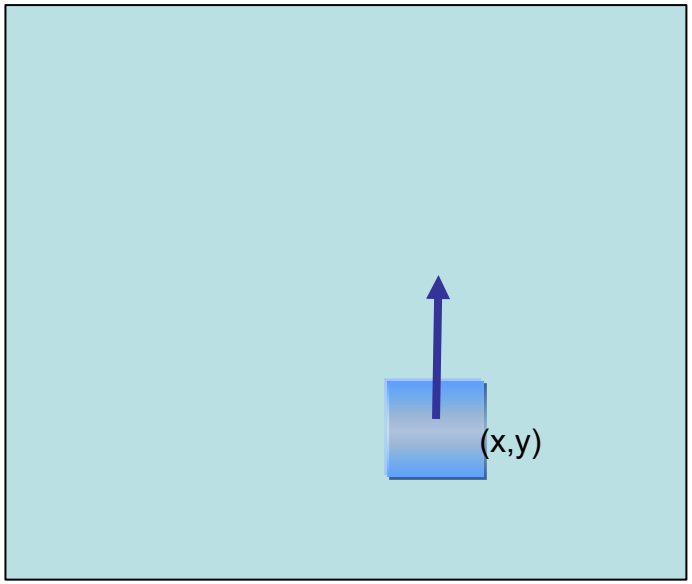
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AUTONOMOUS ROBOT NAVIGATION

OBJECTIVES

- We want a two-wheeled robot that drives autonomously. It should be able to pursue a target and at the same time, avoid the obstacles along the way.





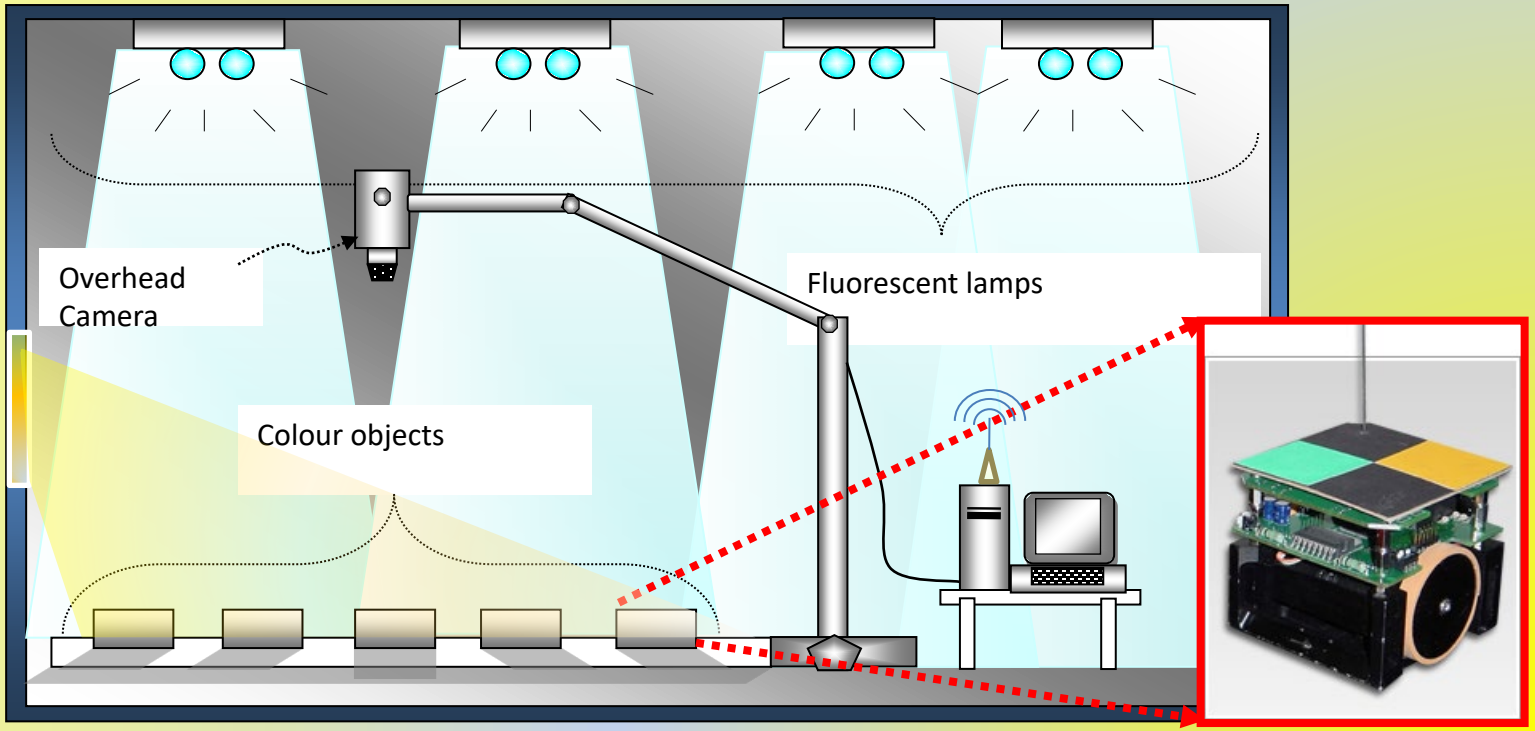
REQUIREMENTS

- Real-time computer vision system
- High-level path planner
- Motor controller
- Communication system

Robot Soccer Game Platform

Compliant to the FIRA International Standards

A problem that has never been solved yet.



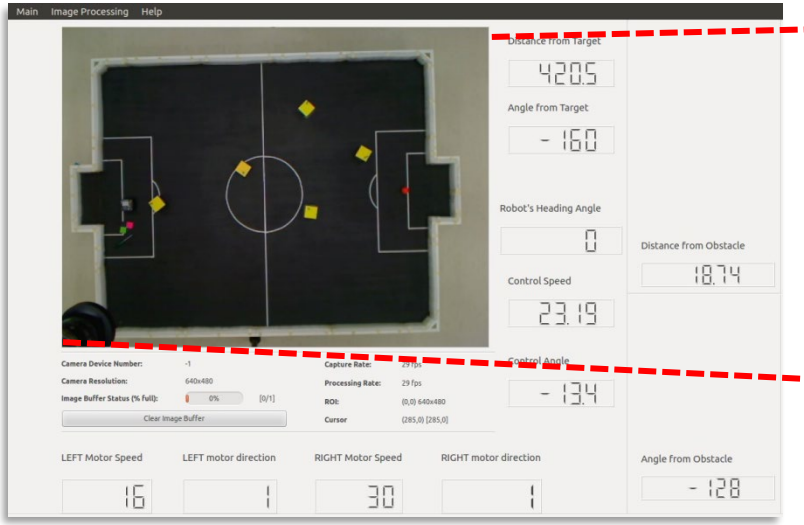
Exploratory environment is indoor – room totally obstructed from sunlight

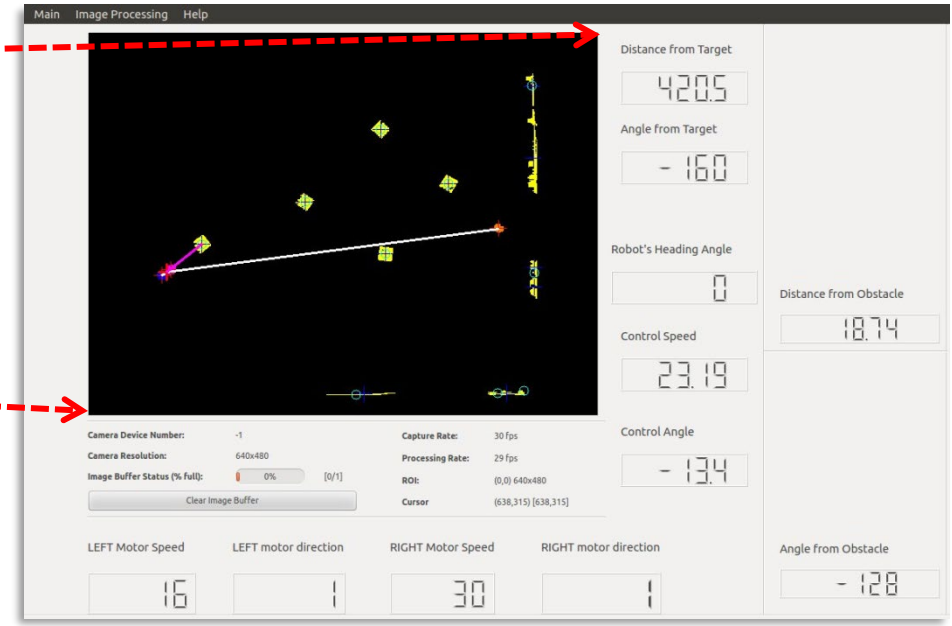
Multiple monochromatic light sources – fluorescent / fluoride lamps

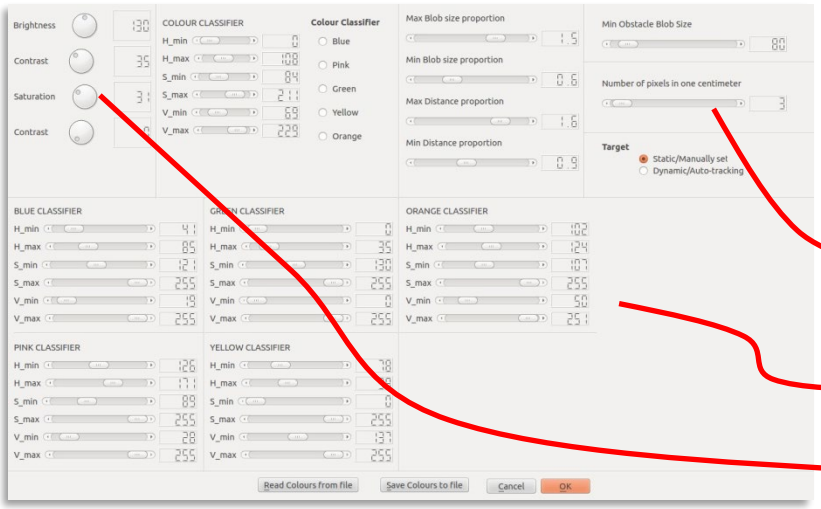
Colour Object Recognition (Recognition speed: ~ 33ms)

Real-time Vision and Control System

Written from scratch, in C/C++ using QT and OpenCV







Pixel-to-centimeter conversion

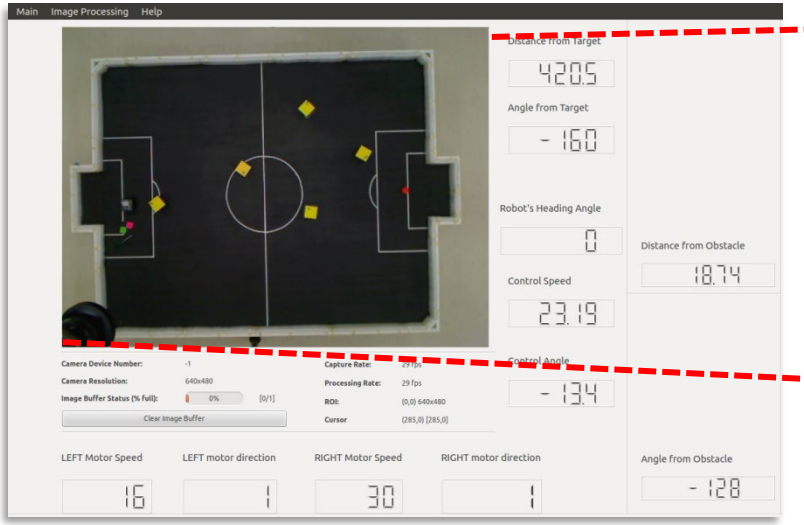
Colour classifiers

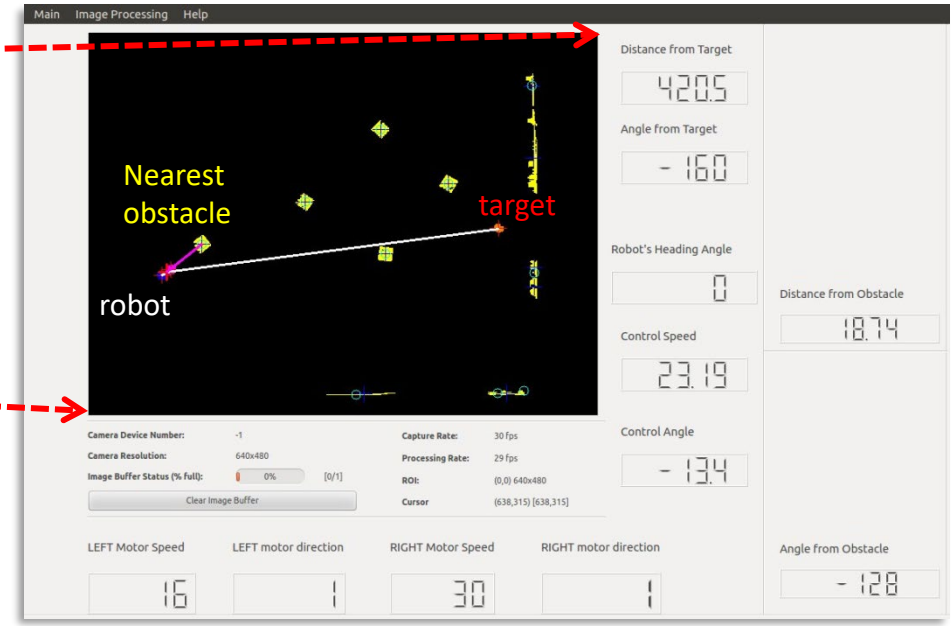
Camera settings

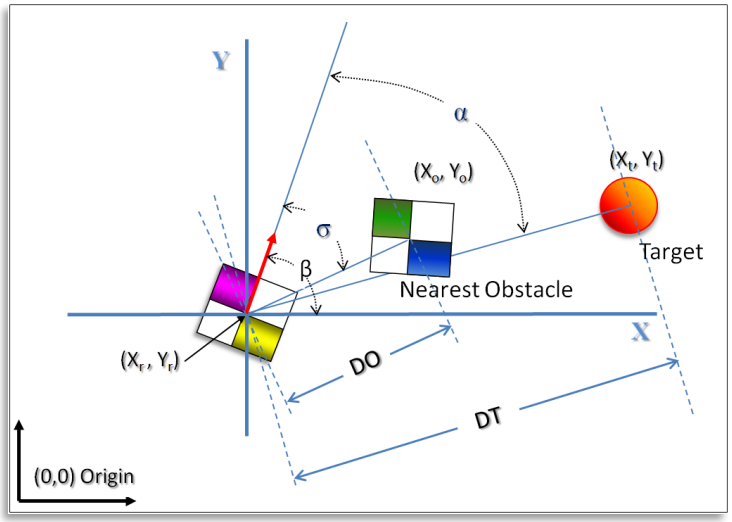


Real-time Vision and Control System

Written from scratch, in C/C++ using QT and OpenCV







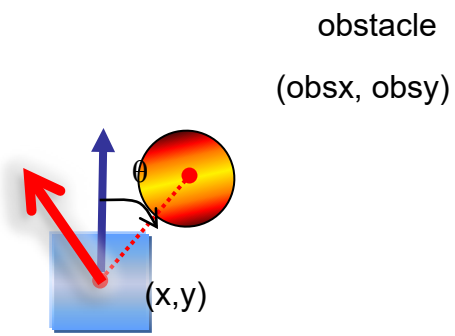
Target:
- angle and distance measurements

nearest obstacle:
- angle and distance measurements

OBSTACLE AVOIDANCE

Consider the case given below, when the robot encounters an obstacle along its path, and that obstacle is found on the right-hand side, relative to the robot, what should the robot do?

Here, it is easy to see that we can simply turn the robot sharply to the left to avoid the obstacle.



Some solutions to a problem could be described linguistically but only very vaguely. Is there an algorithm that could help us translate control rules into its precise mathematical form?

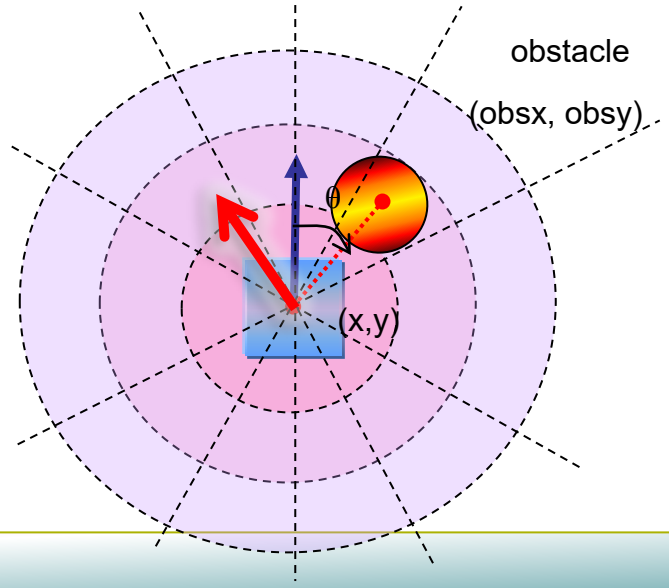


Fuzzy Rule

OBSTACLE AVOIDANCE

If (Distance is NEAR) and (**obstacle is at the right hand side**) and (Angle is SMALL)

Then Turn Sharp Left.



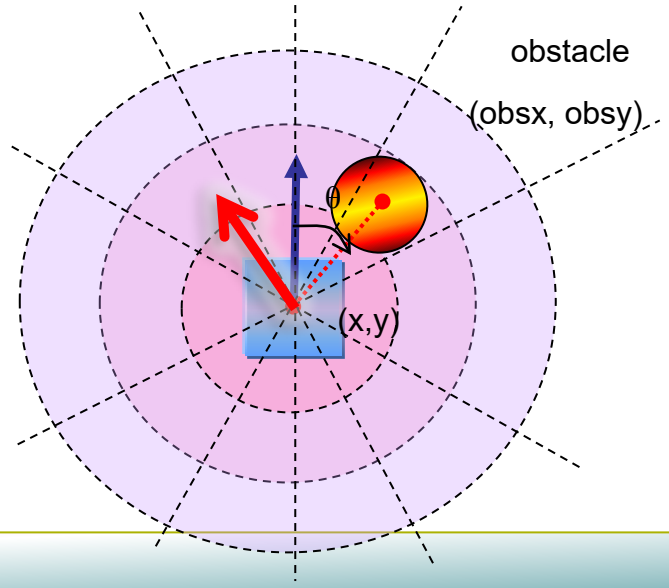
Fuzzy Rule

OBSTACLE AVOIDANCE

If (Distance is NEAR) and (**obstacle is at the right hand side**) and (Angle is SMALL)

Then Turn Sharp Left.

We need a mathematical facility that could translate these linguistic terms into mathematical forms that approximate the actual physical dimensions, and relevant measurements.



What do we mean when we say NEAR or Small or Turn Sharp Left? Humans only make approximate measurements.



Fuzzy Rule

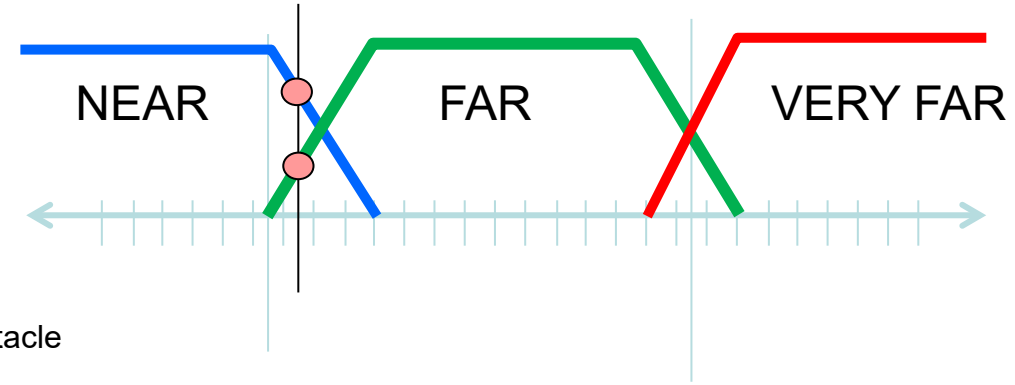
OBSTACLE AVOIDANCE

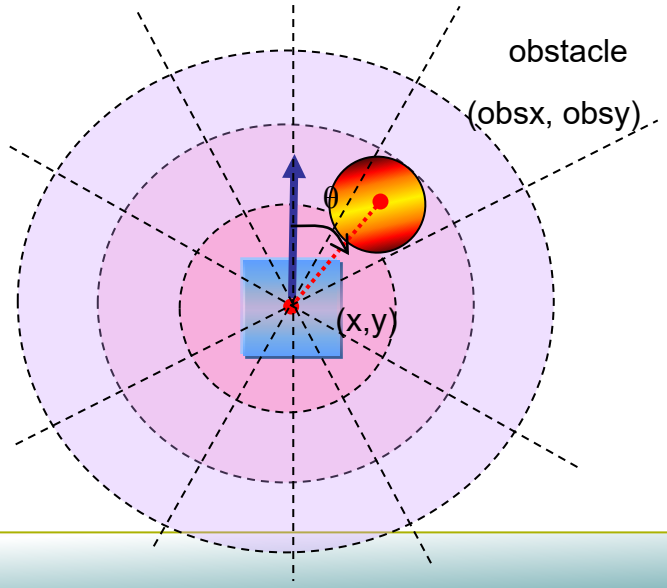
Fuzzy Logic allows (actually encourages) us to define overlapping sub ranges.

It provides all the mathematical functions to quantify our terms and use them in our rules.

If (Distance is NEAR)

…





FUZZY SETS

NEAR: [0, 14.5cm]
FAR: [10.6, 24cm]
VERY FAR: [20cm, +infinity]

Fuzzy Systems for Obstacle Avoidance

Vision System

Nearest Obstacle (Distance and Angle)

Fuzzy System 3 (Steering)

	NEAR	*FAR*	*VERY FAR*
SMALL	Very Sharp	Sharp Turn	Med Turn
MEDIUM	Sharp Turn	Med Turn	Mild Turn
LARGE	Med Turn	Mild Turn	Zero Turn

e.g. If the *Distance* from the Obstacle is *NEAR* and the *Angle* from the Obstacle is *SMALL* **Then** turn *Very Sharply*.

Angle

Fuzzy System 4 (Speed Adjustment)

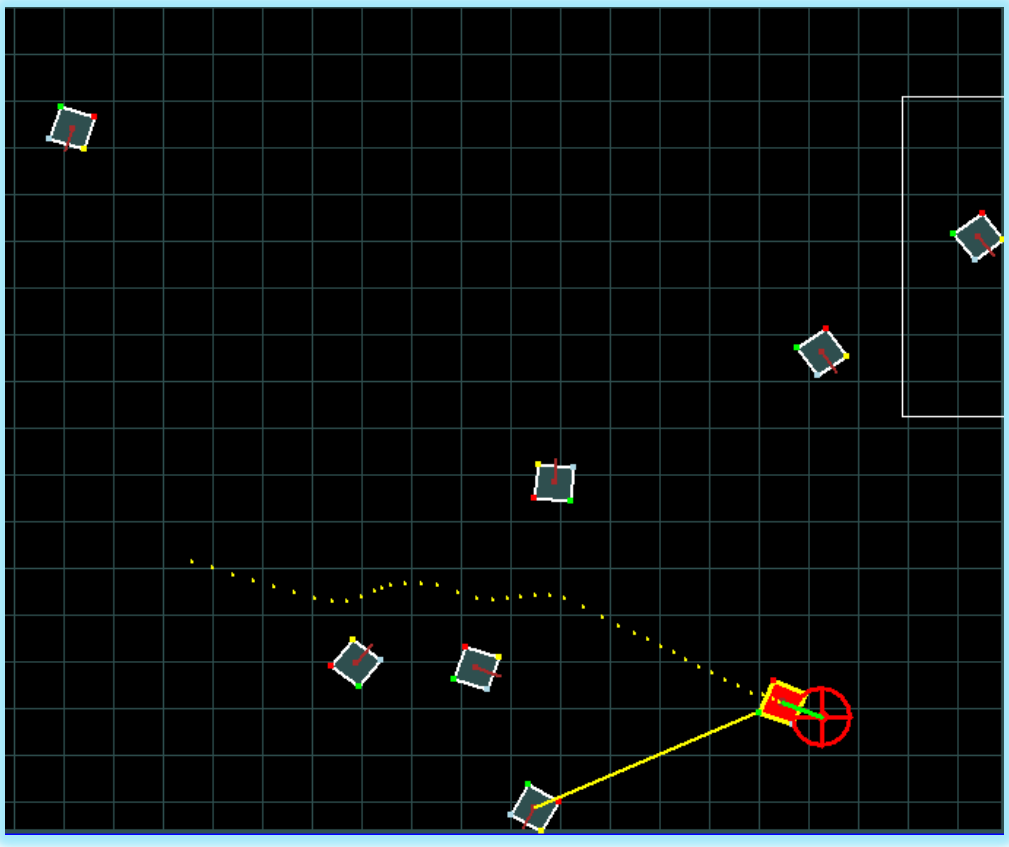
	NEAR	*FAR*	*VERY FAR*
SMALL	Very Slow	Slow Speed	Fast Fast
MEDIUM	Slow Speed	Fast Speed	Very Fast
LARGE	Fast Speed	Very Fast	Top Speed

e.g. **If** the *Distance* from the Obstacle is *NEAR* and the *Angle* from the Obstacle is *SMALL* **Then** move *Very Slowly.*

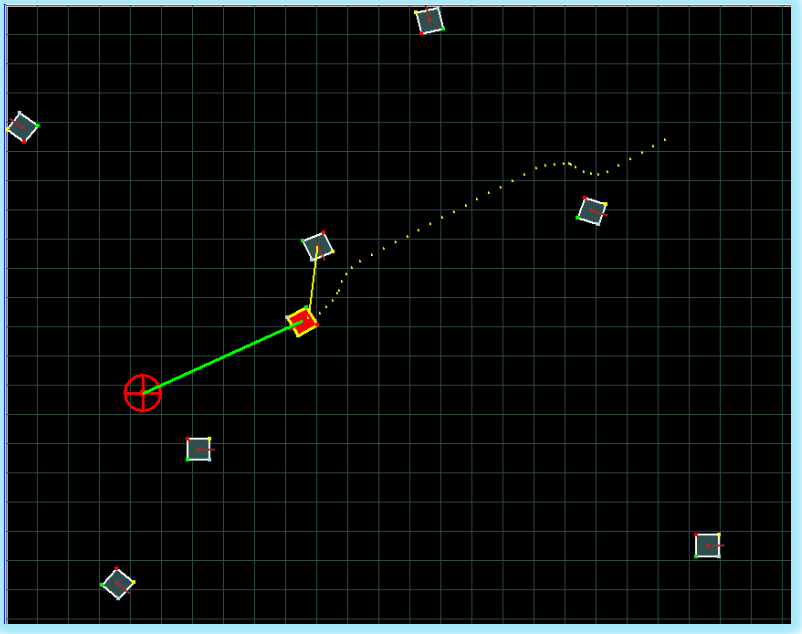
Speed

Smooth path-planning in real-time

Cascade of Fuzzy Logic Systems



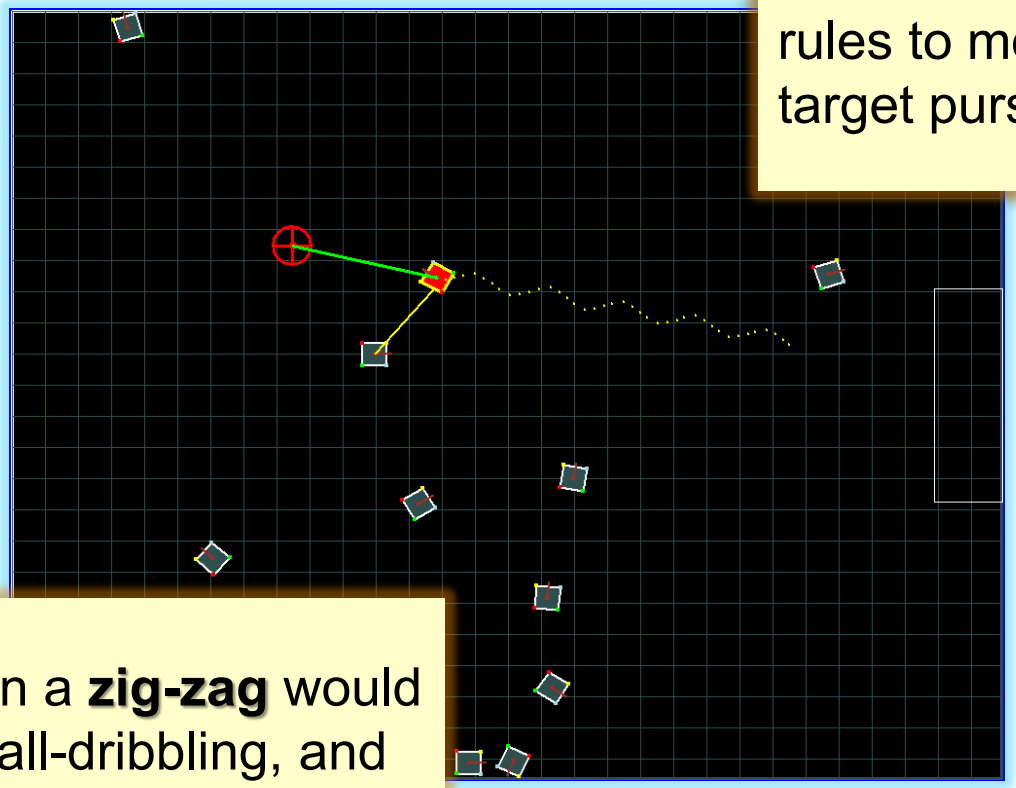
Combination of target pursuit and obstacle avoidance behaviours



Drunken-robot behaviour

Move like a fly, sting like a bee!

We can tweak the fuzzy rules to modify the robot's target pursuit behaviour.



A robot moving in a **zig-zag** would be suitable for ball-dribbling, and attacking the goal, for instance.

Let's see the simulation.

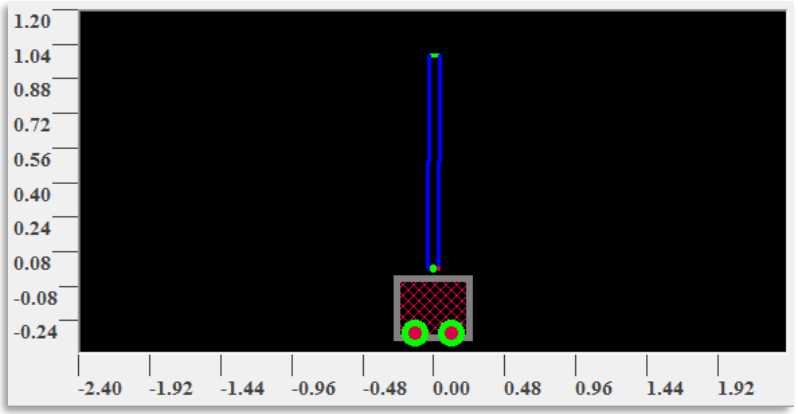
Other robot behaviours

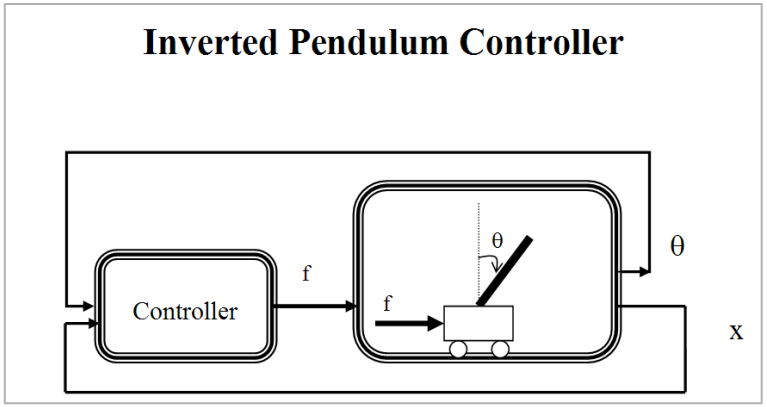
- **Auto parking**
- **Opponent evasion**
- **Wall avoidance**
- **Wall following**
- **Ball dribbling**
- **Ball shooting**
- **Ball passing**
- **Goal keeper intelligence**
- **Team coach**

- Many robot behaviours can be developed using the same design strategy.
- Various intelligent systems could be fused together to accomplish this. But integrating them together may be tricky.

Let's see the video of robot soccer using fuzzy logic

Inverted Pendulum Problem





1. **Input variables (state vector):** sensed and derived values

   Sensed values:

   a) **x** – position of object with respect to the horizontal axis

   b) **θ** - angle of pole relative to the vertical axis

   Derived values:

   c) **x'** - velocity along the x-axis

   d) **θ'** - angular velocity

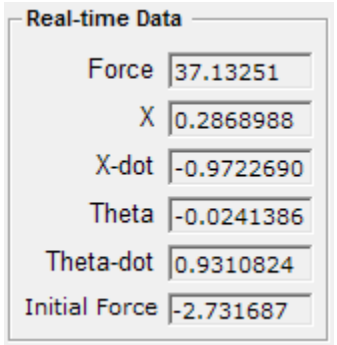
2. **Controller output:** **F** – force to be applied to the cart

menu

Fuzzy Logic Controlled Inverted Pendulum

$$\theta''(t) = \frac{mg\sin(\theta(t)) - \cos(\theta(t))[F(t) + m_b l(\theta'(t))^2 \sin(\theta(t))]}{(4/3)ml - m_b l\cos^2(\theta(t))}$$

$$x''(t) = \{F(t) + m_b l(\theta'(t))^2 \sin(\theta(t)) - \theta'(t)\cos(\theta(t))]\} / m$$



$\theta(t)$ = broom angle(with vertical) at time t (in radians)

$\theta'(t)$ = angular velocity

x(t) = cart position at time t (in meters)

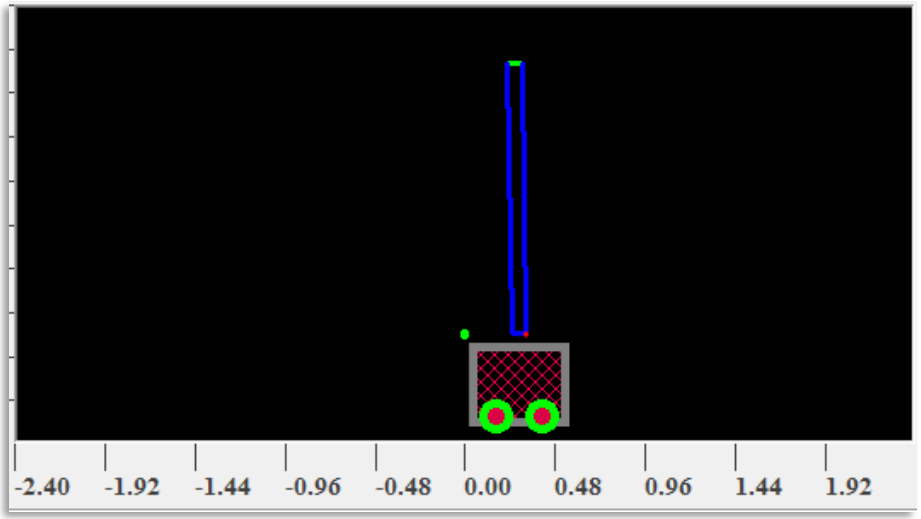
x'(t) = cart velocity

F(t) = force applied to cart at time t (Newtons)

m = combined mass of cart and broom (1.1 kg)

m_b = mass of broom (0.1 kg)

l = length of broom (pivot point to center of mass, 0.5 meters)



menu

Search

- Systematic exploration of alternatives
- Find a path
- Find a plan of action

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Search Algorithms

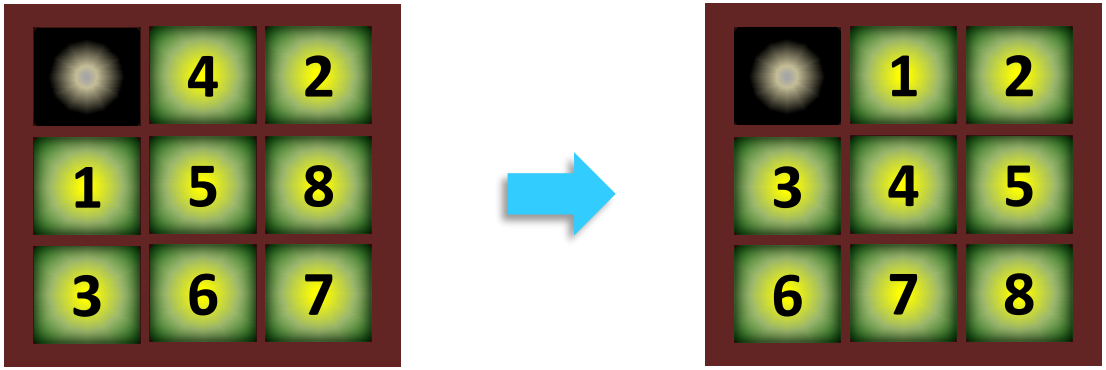
Any-Path vs. Optimal Path

UNINFORMED vs. INFORMED

COMPLETE vs. INCREMENTAL

8 Puzzle Problem

Given the **start** state, find the sequence of moves that will arrange the tiles according to the **goal** state.



Start **Goal**

Tiles can be moved **up, down, left,** or **right**, provided that the following conditions are met:

- There is no other tile blocking the direction of the movement.
- Tiles are not moved outside of the boundaries.

8 Puzzle Problem

Given the **start** state, find the sequence of moves that will arrange the tiles according to the **goal** state.

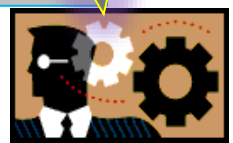


Start **Goal**

Wouldn't be nice to learn an algorithm that can solve puzzles really quickly and always gives us the best solution?

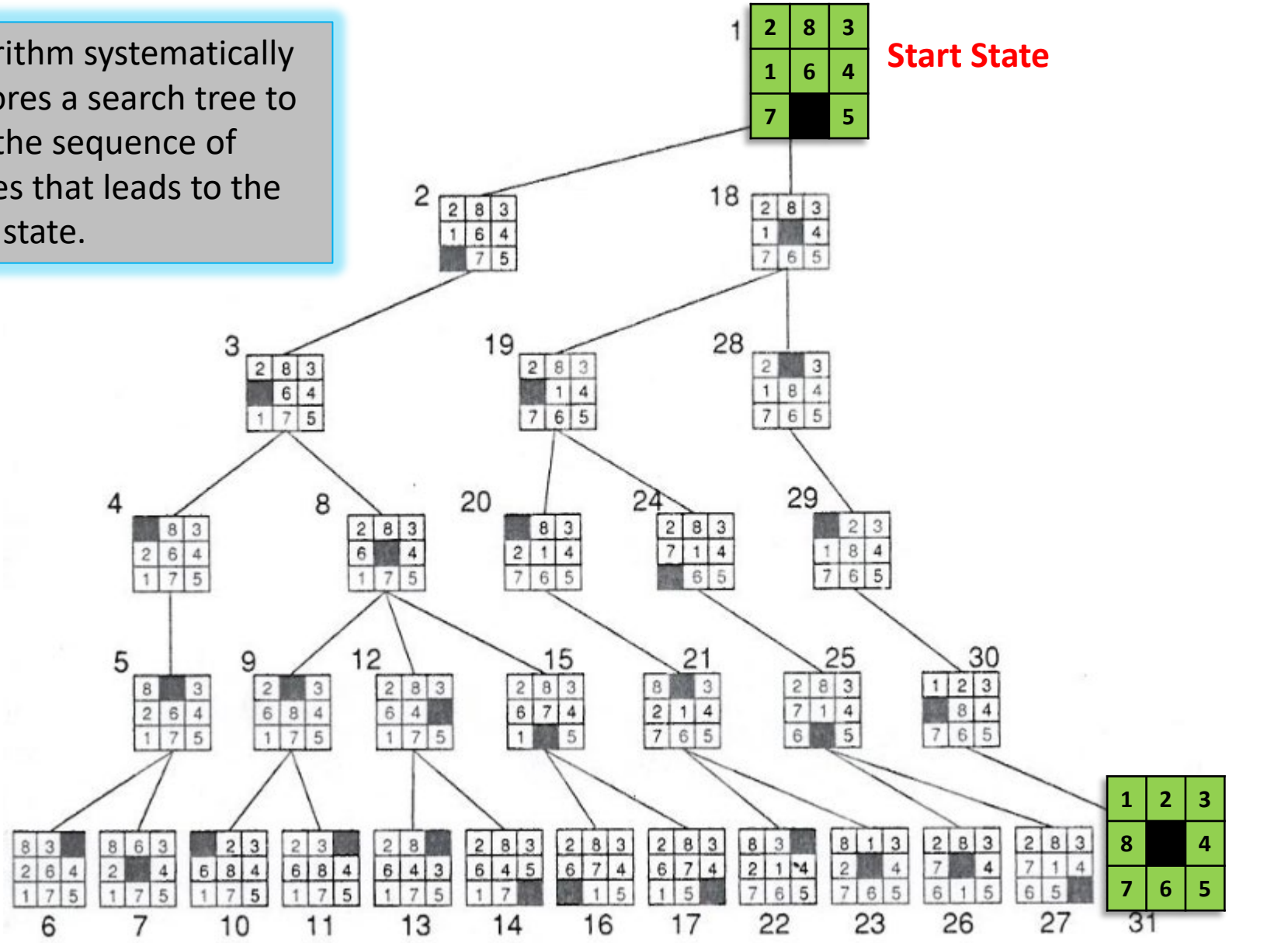
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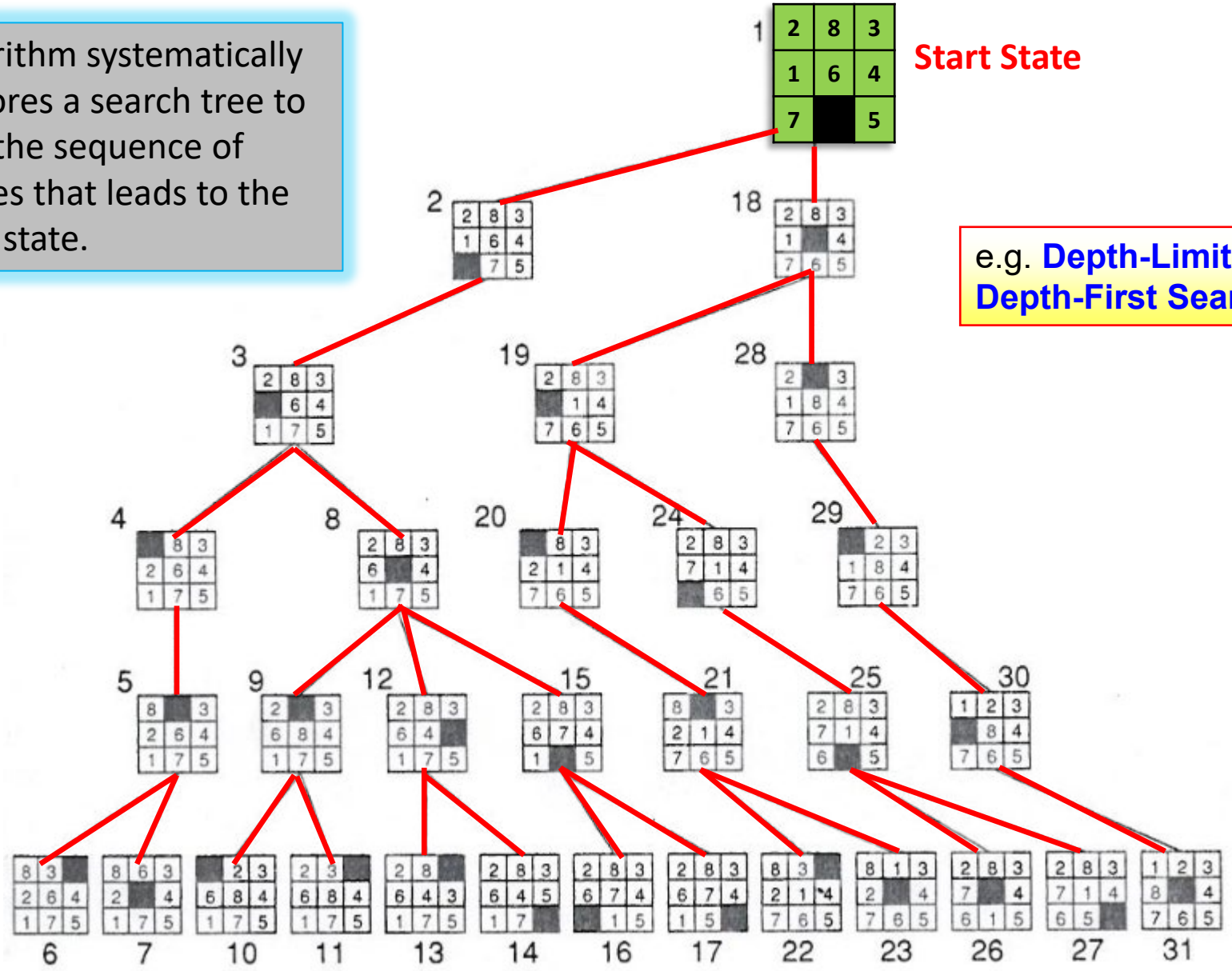
Search

Algorithm systematically explores a search tree to find the sequence of moves that leads to the Goal state.



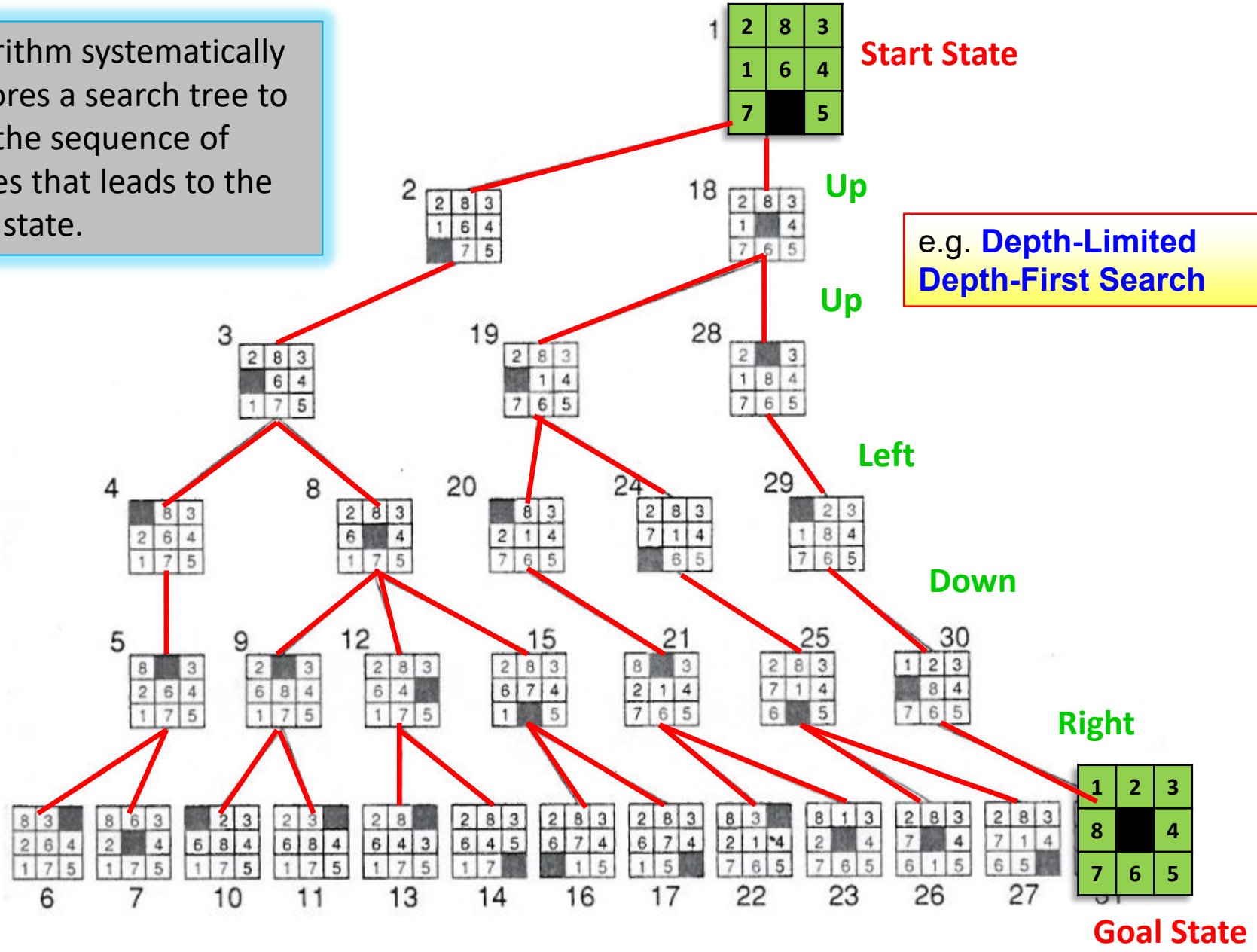
Uninformed Any-Path Search

Algorithm systematically explores a search tree to find the sequence of moves that leads to the Goal state.



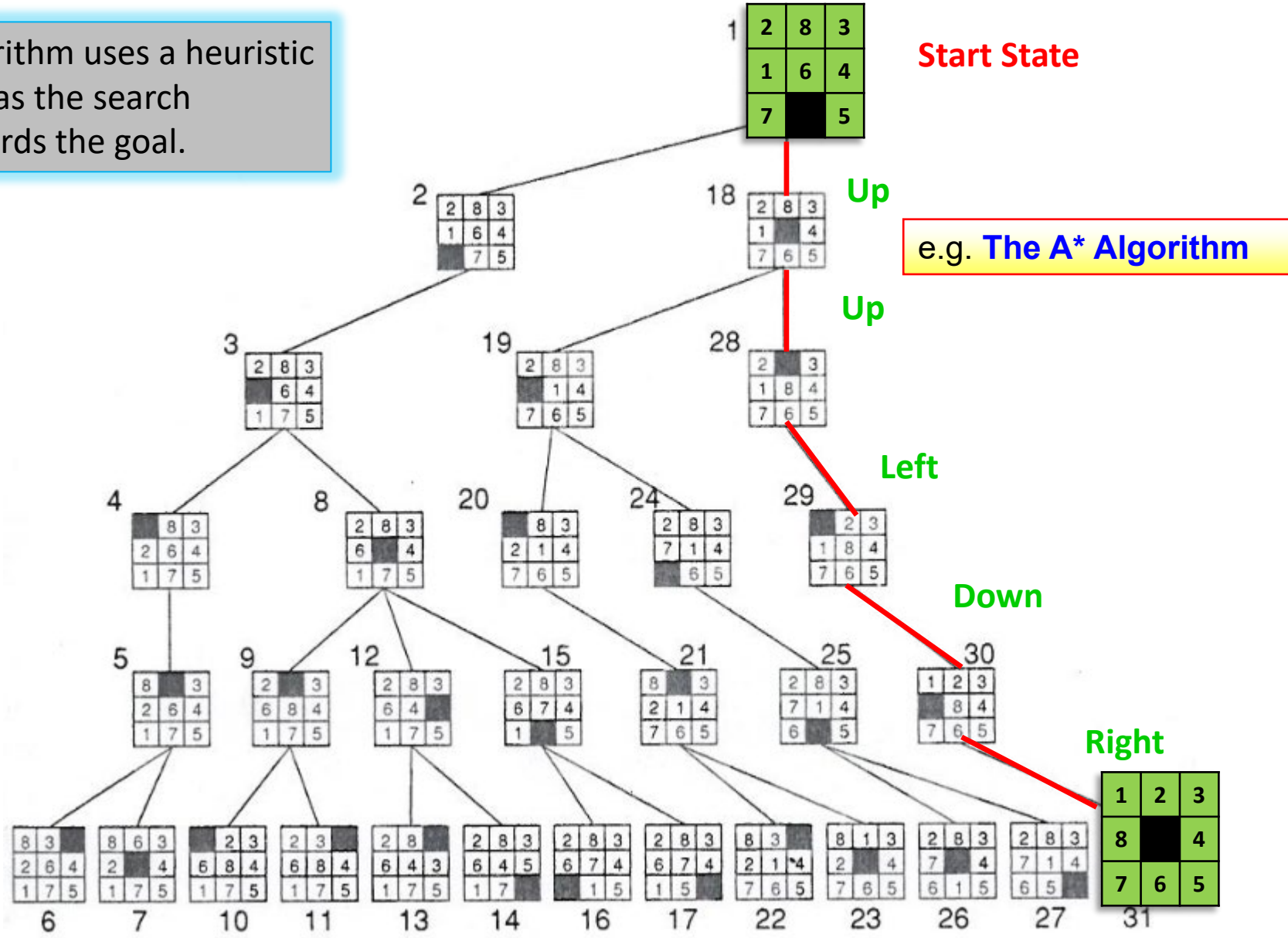
Uninformed Any-Path Search

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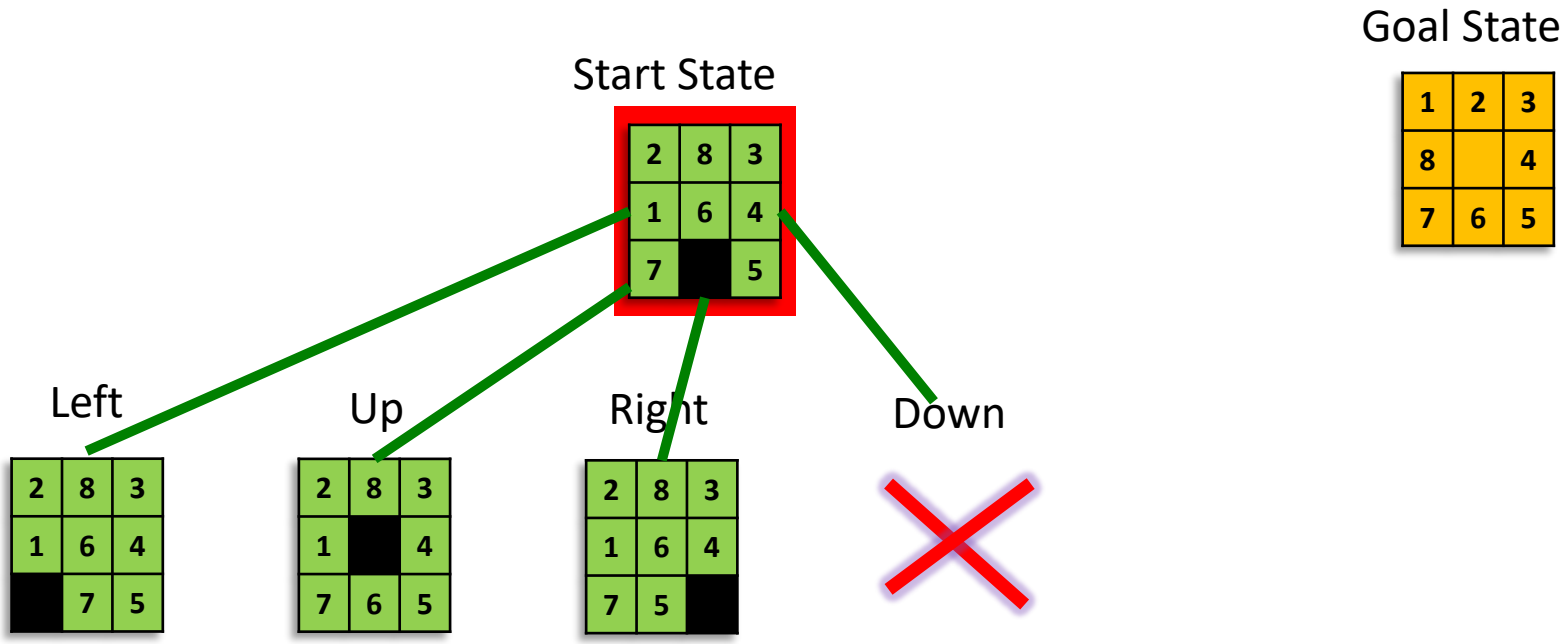


Informed, Optimal Search

Algorithm uses a heuristic to bias the search towards the goal.

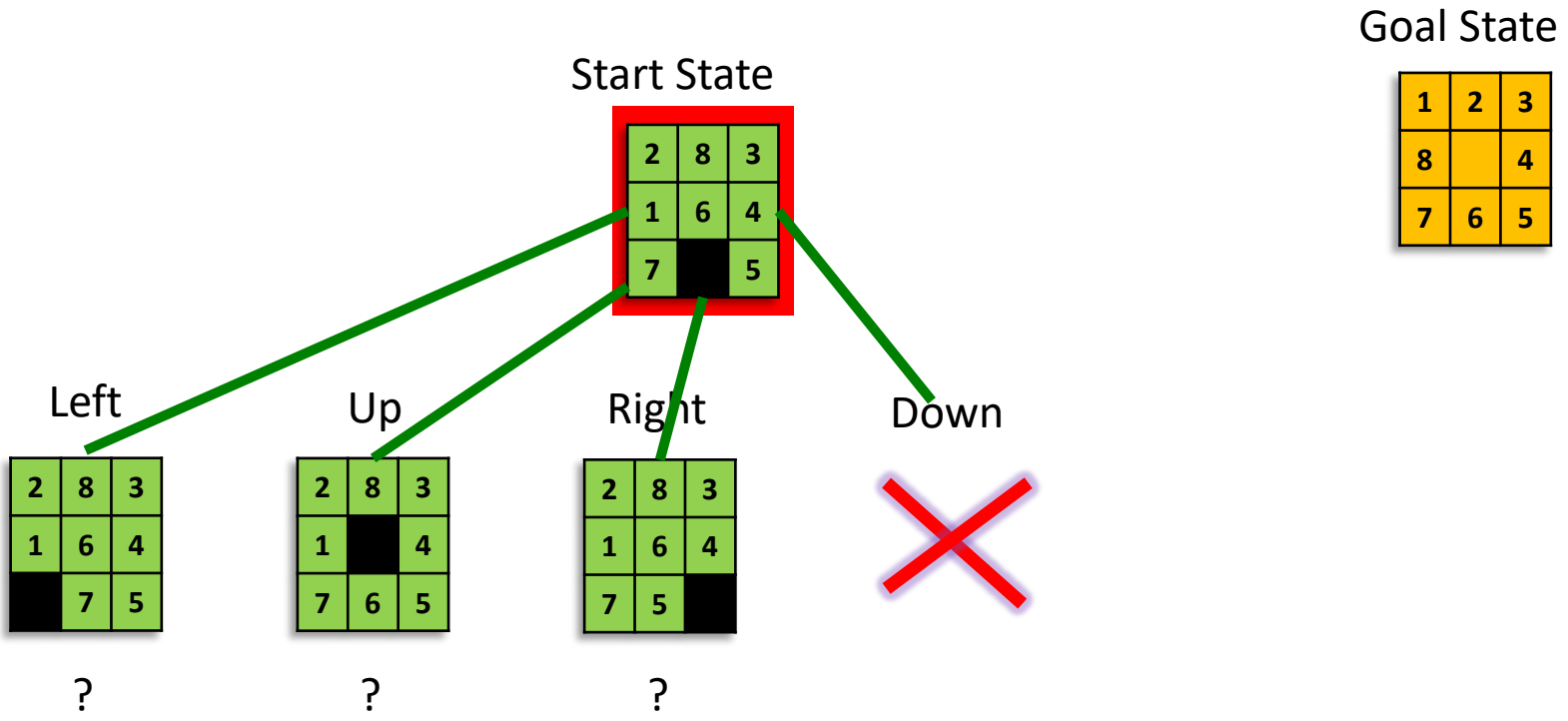


Search



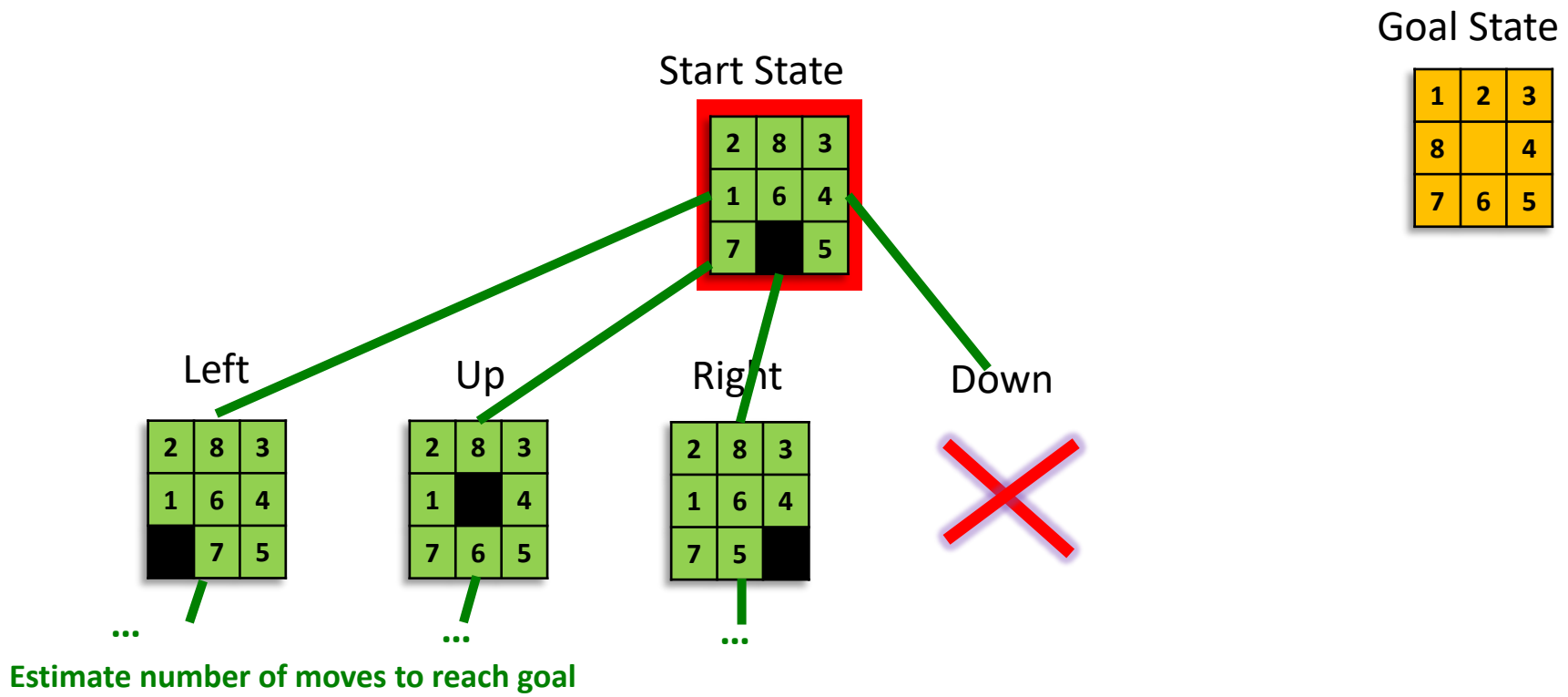
Given the start state, we can examine the different possible moves that can be carried out from that state.

Search



But can we tell which among the tiles will likely lead us to the goal (quickly!) ?

Heuristic Search



But can we tell which among the tiles will likely lead us to the goal (quickly!) ?

We will see that we can use a heuristic function to find an estimate of the remaining number of moves to reach the goal.

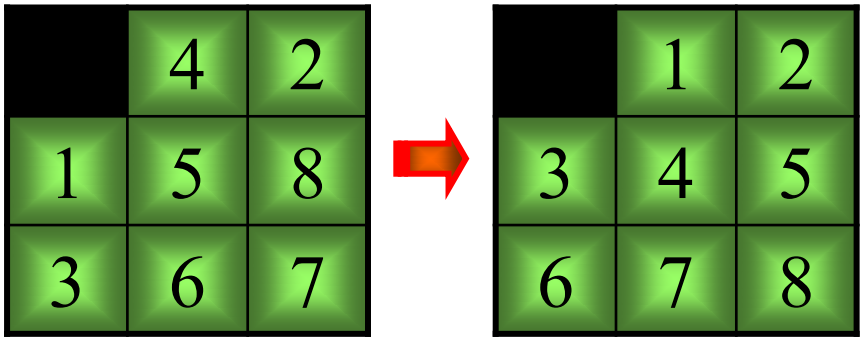
Sum of Manhattan Distance of tile to its goal location

Manhattan Distance between (x_1, y_1) and (x_2, y_2) is $|x_1 - x_2|$ and $|y_1 - y_2|$.

Each move can only decrease the distance of exactly one tile.

Using search algorithms to solve the 8 Puzzle Problem

8 Puzzle: Move tiles to reach goal. Think of a move as moving an "empty" tile.



S G

Let us see some demo…